

Percent of Change: Sales Tax, Discounts, Markup & Reverse Problems

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Standard: CCSS.MATH.CONTENT.7.RP.A.3 — Solve multistep ratio and percent problems including percent increase/decrease, simple interest, tax, markups, markdowns, gratuities, commissions, fees, and finding the percent of change.

Quick Reference — From Today's Lecture

- **Percent of Change formula:** $\frac{\text{Amount of Change}}{\text{Original Amount}} = \frac{X}{100}$
- **Sales Tax:** Tax = Rate $\times$ Marked Price; Total Cost = Marked Price + Tax
- **Discount:** Discount = Rate $\times$ List Price; Sale Price = List Price – Discount
- **Markup (cost $\rightarrow$ selling price):** Markup % = (Selling Price – Cost) $\div$ Cost $\times$ 100%
- **Reverse:** Original = Sale Price $\div$ (1 – rate); Original = Total $\div$ (1 + rate)

Part A — Sales Tax and Total Cost

MEDIUM

Directions: Find the sales tax and the total cost.

1. A radio is marked at \$50, with a 6% sales tax rate. Find the sales tax and the total cost.

Sales tax: \$ _____ Total cost: \$ _____

2. A bookshelf is marked at \$200, with an 8% sales tax rate. Find the sales tax and the total cost.

Sales tax: \$ _____ Total cost: \$ _____

Directions: Find the discount and the sale price.

3. A sweater has a list price of \$80, and the rate of discount is 25%. Find the discount and the sale price.

Discount: \$ _____ Sale price: \$ _____

4. A jacket has a list price of \$150, and the rate of discount is 40%. Find the discount and the sale price.

Discount: \$ _____ Sale price: \$ _____

Part C — Percent of Change (Increase or Decrease)

MEDIUM

Directions: Identify whether the change is an increase or a decrease, then find the percent of change. Use

$$\frac{\text{change}}{\text{original}} = \frac{X}{100}.$$

5. The number of points went from 40 to 60. Identify the change and find the percent of change.

_____ of _____ %

6. A price went from 80 to 60. Identify the change and find the percent of change.

_____ of _____ %

7. A score went from 25 points to 50 points. Identify the change and find the percent of change.

_____ of _____ %

8. The number of party invitations went from 125 to 75. Identify the change and find the percent of change.

_____ of _____ %

Part D — Find the Original Amount (Reverse)

MEDIUM

Directions: *Work backward to find the price before the discount or tax was applied.*

9. After 20% off, a pair of shoes sells for \$40. What was the original price?

Original price: \$ _____

10. Including a 10% sales tax, a hat costs \$44. What was the price before tax?

Original price: \$ _____

▼ CHALLENGING SECTION ▼

Part E — Markup (Cost → Selling Price)

CHALLENGE

Directions: *Find the percent of markup using $\text{markup \%} = \frac{\text{Selling Price} - \text{Cost}}{\text{Cost}} \times 100\%$.*

11. A pair of Nike sneakers costs Nike only 20 to make, but customers pay 150 to buy them. What is the percent of markup?

Markup: _____ %

12. A store buys a backpack from a supplier for 40 and sells it to customers for 100. What is the percent of markup?

Markup: _____ %

Part F — Find the Rate

CHALLENGE

Directions: Use $\frac{\text{change}}{\text{original}} = \frac{X}{100}$ to find the missing percent.

- 13.** A shirt is marked at 40, and the total cost (with sales tax) comes out to 42. What is the sales tax rate?

Tax rate: _____ %

- 14.** A bicycle has a list price of 120, and the sale price is 90. What is the rate of discount?

Discount rate: _____ %

Part G — Multi-Step Word Problems

CHALLENGE

Directions: Read carefully. Apply each step in the right order.

- 15.** A bicycle is originally priced at \$200. The store first applies a 25% discount, then adds a 6% sales tax to the discounted price. Find the final total cost.

Final total: \$ _____

- 16.** A guitar is originally priced at \$300. The store first applies a 30% discount, then adds a 10% sales tax to the discounted price. Find the final total cost.

Final total: \$ _____

17. You buy a book that is on sale for 30% off, and you pay \$63. Your friend wants to know the original price of the book. Find the original price.

Original price: \$ _____

18. One week, 72 people got a speeding ticket on Main Street. The next week, only 36 people got a speeding ticket. What is the percent of change in speeding tickets?

_____ of _____ %

19. The same pair of sneakers is on sale at two stores. Store A regular price is 50, with 30% off. Store B regular price is 60, with 40% off. Which store has the lower sale price, and by how much?

Lower store: Store _____ Difference: \$ _____

20. A jacket is originally priced at \$200. It is on sale for 25% off, and you also have a coupon that takes an extra 10% off the already-discounted sale price. Find the final price you actually pay.

Final price: \$ _____